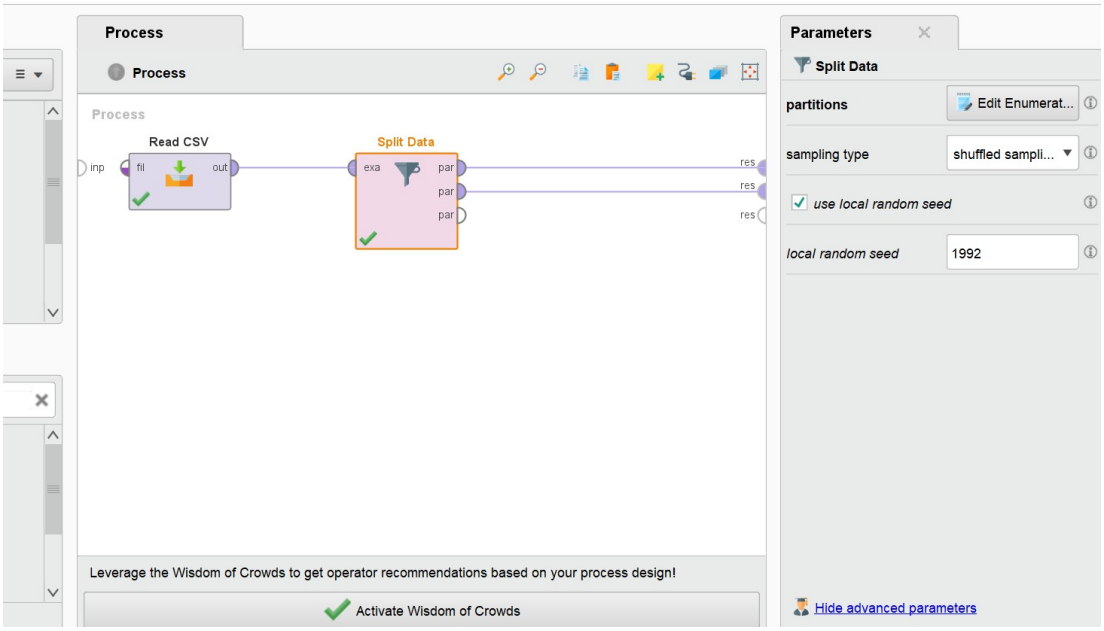
Predicting Haaland's Goals in Altair AI Studio

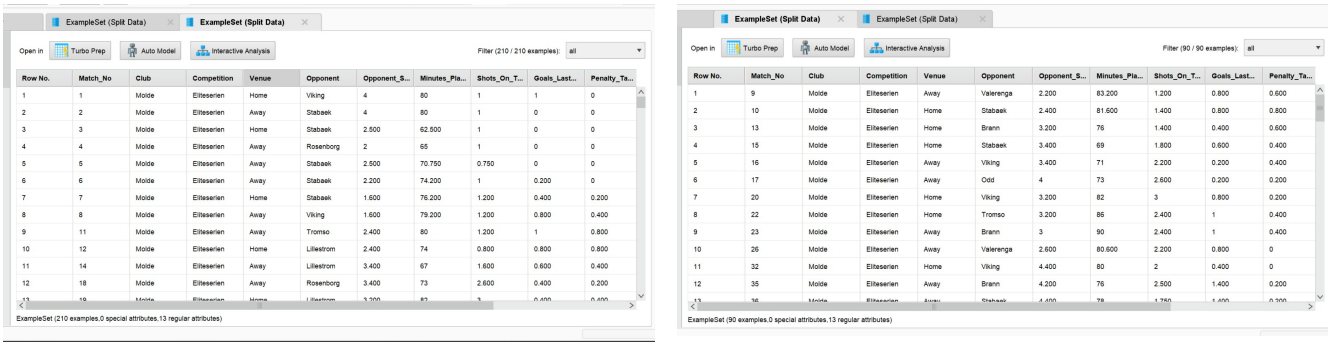
Goal: predict the **Goals** column from the other (known) columns. Steps: split the data, build a Decision Tree, then a Random Forest.

Step 1: Split the Data

The data was imported with **Read CSV** and connected to **Split Data**. Its parameters use **shuffled sampling** (random rows) and two partitions set with Edit Enumeration, which divide the 300 matches into **70% training data (210 matches)** and **30% testing data (90 matches)**. The local random seed (1992) keeps the same split every time it runs.



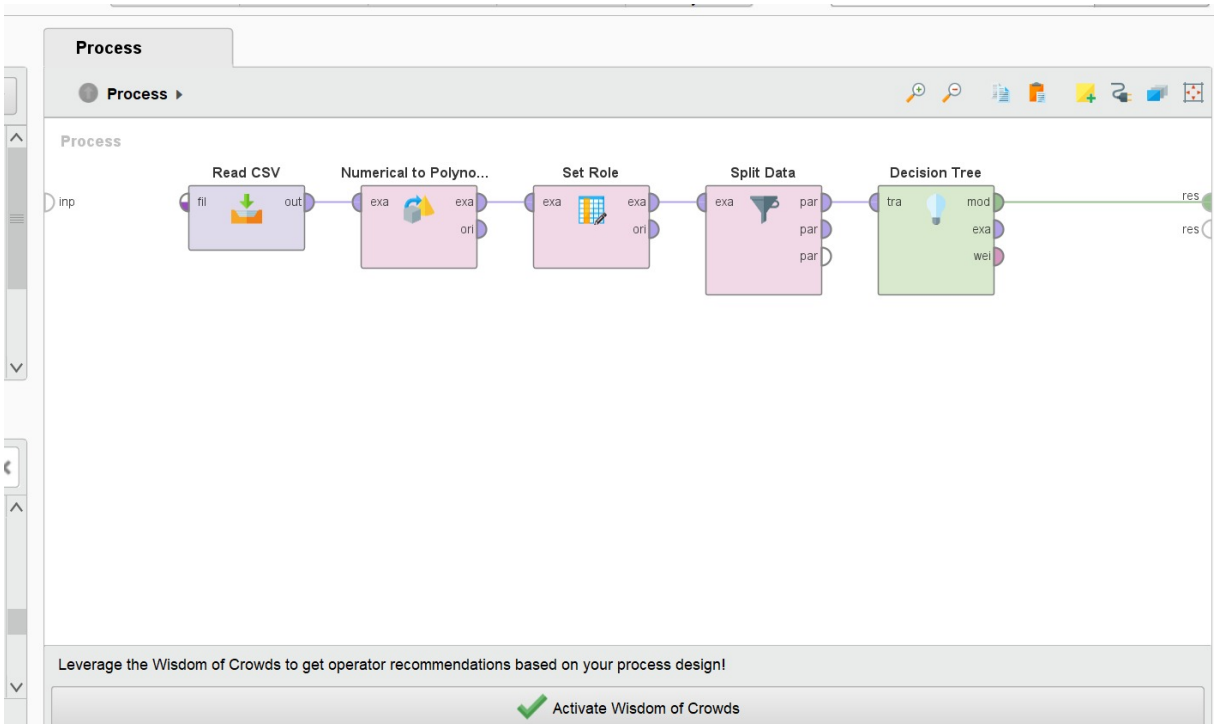
Split Data design and its parameters.



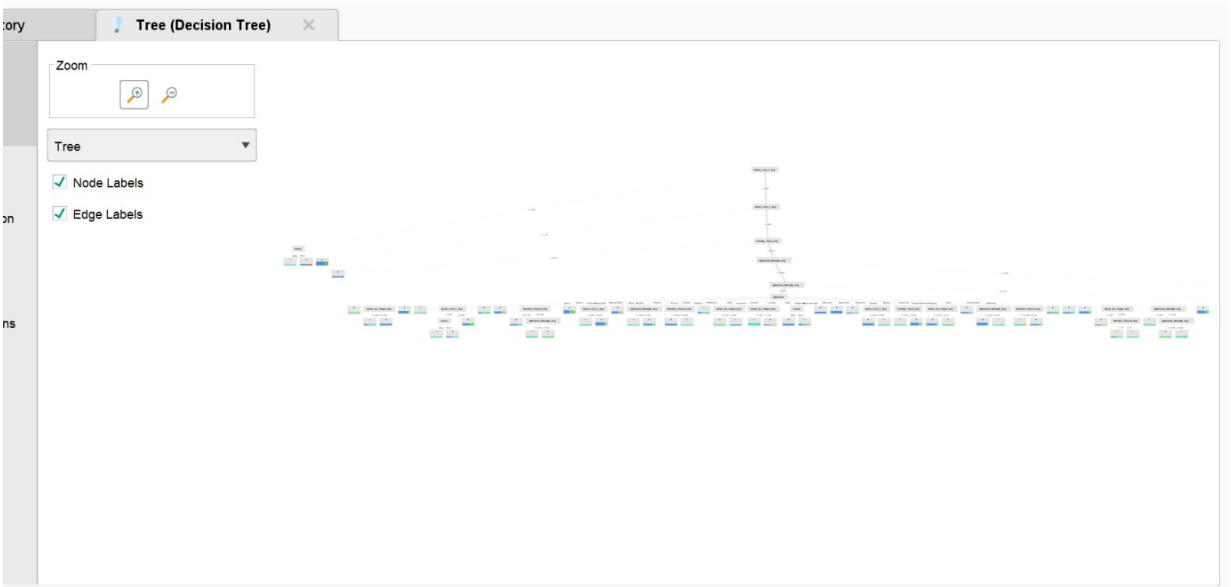
Left: training data (210 examples). Right: testing data (90 examples).

Step 2: Decision Tree

Numerical to Polynominal turns Goals into classes (0, 1, 2...) and **Set Role** marks Goals as the label (what we predict). The training part (70%) from Split Data is then connected to the **Decision Tree** operator, and its model output is sent to the results. The tree learns rules from columns such as Goals_Last_5_Avg, Shots_On_Target_Avg and Opponent_Strength_Avg to decide how many goals Haaland is likely to score.



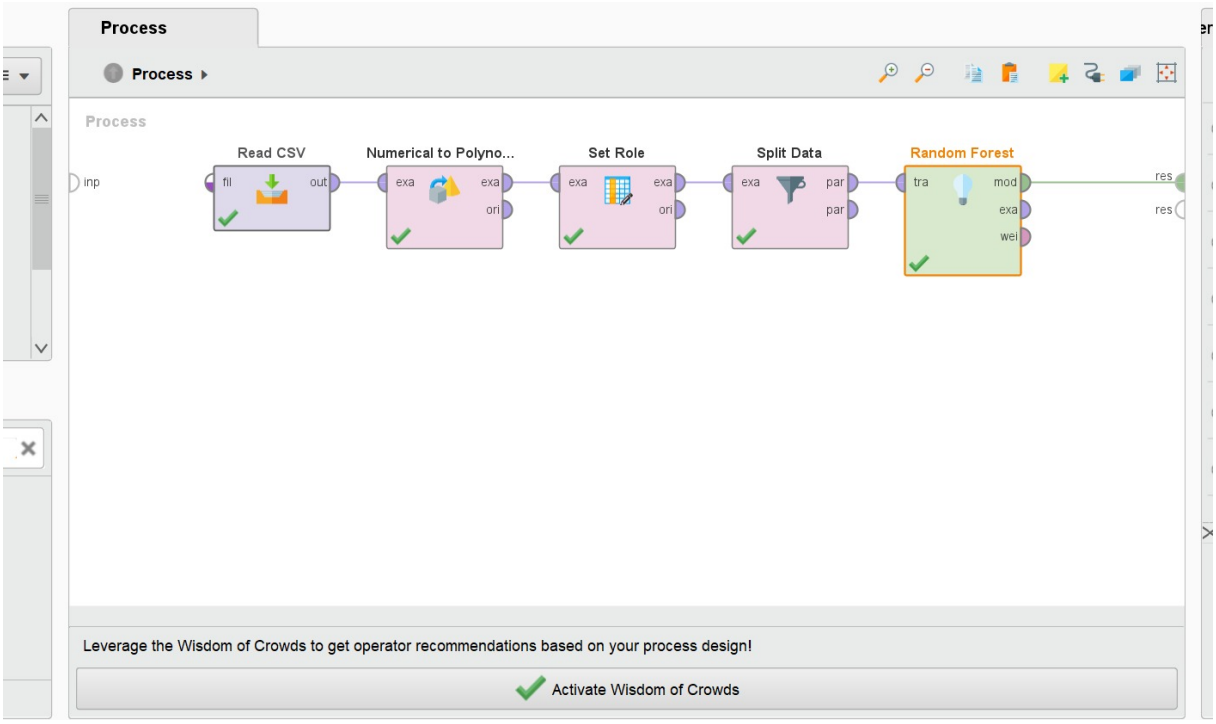
Process design: Read CSV, Numerical to Polynominal, Set Role, Split Data, Decision Tree.



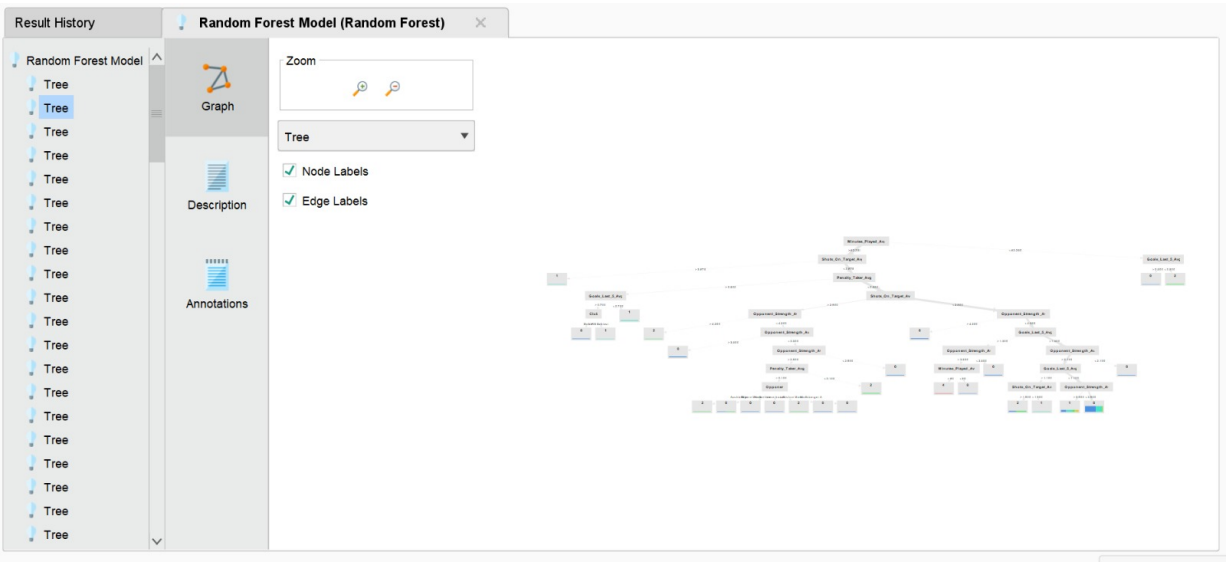
Resulting decision tree (Graph view). Each box is a question about a column and each end box is a predicted Goals value.

Step 3: Random Forest

Random Forest replaces the Decision Tree in the same process. Instead of one tree it builds many trees, each from a random part of the training data, and combines their results. This is usually more reliable than a single tree.



Process design with Random Forest connected to the training part of Split Data.



Random Forest model: the list on the left shows the individual trees, and the graph shows one selected tree.

Summary: Haaland's data was split 70/30 (210 train, 90 test), then a Decision Tree (one tree) and a Random Forest (many trees) were trained to predict Goals.